

```

clc
clear all
format short
% To solve the LPP by Simplex Method
%Min z=x1-3x2+2x3
%Subject to 3x1-x2+2x3<=7
%-2x1+4x2<=12
%-4x1+3x2+8x3<=10
%x1,x2,x3>=0
%First to change the objective function from minimization to maximization
%Max z=-x1+3x2-2x3
%To input Parameters
C=[-1 3 -2];
info=[3 -1 2; -2 4 0; -4 3 8];
b=[7; 12; 10];
NOVariables=size(info,2);
s=eye(size(info,1));
A=[info s b];
Cost=zeros(1,size(A,2));
Cost(1:NOVariables)=C;
%Constraint Basic Variable
BV=NOVariables+1:size(A,2)-1;
%calculate Zj-Cj row
ZRow=Cost(BV)*A-Cost;
%To print the table
ZjCj=[ZRow;A];
SimpTable=array2table(ZjCj);
SimpTable.Properties.VariableNames(1:size(ZjCj,2)) =
{'x_1','x_2','x_3','s_1','s_2','s_3','Sol'}
%Simplex Table starts
Run=true;
while Run
if any(ZRow<0) %To check any negative value is there
    fprintf('The current BFS is not Optimal \n')
    fprintf('\n =====The Next Iteration Results=====\n')
    disp('Old Basic Variable (BV)=')
    disp(BV)
    %To find the entering variable
    ZC=ZRow(1:end-1);
    [EnterCol,Pvt_Col]=min(ZC);
    fprintf('The most negative element in ZRow is %d Corresponding to Column %d \n',EnterCol, Pvt_Col)
    EnterCol, Pvt_Col)
    fprintf('Entering Variable is %d \n', Pvt_Col)
    %To find the leaving variable
    sol=A(:,end);
    Column=A(:,Pvt_Col);
    if all(Column<=0)
        error('LPP has unbounded solution. All entries <=0 in Column %d', Pvt_Col)
    else

        %To check minimum ration is with positive entering column entries
        for i=1:size(Column,1)
            if Column(i)>0
                ratio(i)=sol(i)./Column(i);
            else

```

```

        ratio(i)=inf;
    end
end
    %To Finding the minimum Ratio
    [MinRatio, Pvt_Row]=min(ratio);
    fprintf('Minimum ratio corresponding to pivot row is %d \n',Pvt_Row)
    fprintf('Leaving Variable is %d \n', BV(Pvt_Row))
    end
    BV(Pvt_Row)=Pvt_Col;
    disp('New Basic Variables (BV) =')
    disp(BV)
    %Pivot Key
    Pvt_Key=A(Pvt_Row,Pvt_Col);
    %Update the Table for next Iteration
    A(Pvt_Row,:)=A(Pvt_Row,:)./Pvt_Key;
    for i=1:size(A,1)
        if i~=Pvt_Row
            A(i,:)=A(i,:)-A(i,Pvt_Col).*A(Pvt_Row,:);
        end
        ZRow=ZRow-ZRow(Pvt_Col).*A(Pvt_Row,:);
        %To print the table
ZjCj=[ZRow;A];
SimpTable=array2table(ZjCj);
SimpTable.Properties.VariableNames(1:size(ZjCj,2)) ←
{'x_1','x_2','x_3','s_1','s_2','s_3','Sol'}

BFS=zeros(1,size(A,2));
BFS(BV)=A(:,end);
BFS(end)=sum(BFS.*Cost);
CurrentBFS=array2table(BFS);
CurrentBFS.Properties.VariableNames(1:size(CurrentBFS,2)) ←
{'x_1','x_2','x_3','s_1','s_2','s_3','Sol'}
    end

else
    Run=false;
    fprintf('=====\n')
    fprintf('The current BFS is optimal and Optimality is reached \n')
    fprintf('=====\n')
end
end

```